

SQ-OS · WORKING NAME · COMPLIANCE & QUALITY OPERATING SYSTEM

The compliance & quality OS for automotive's long tail

Where every Tier 2–3 supplier proves quality — PPAP, FMEA, traceability — without the spreadsheets that get them rejected.

THE THESIS An affordable, supplier-native compliance & quality OS for the automotive **Tier 2–3 long tail**, with a part-level **traceability / battery-passport backbone** as its data spine. Own the boring, mandated, mission-critical middle of the supply chain — the layer nobody serves affordably.

01 Why now — a margin-crushed, mandate-loaded supply base

4.7%**Average supplier EBIT margin, 2024**

Down from 5.3% in 2023 — near a historic low.

✓ VERIFIED

>40%**Of the 25 largest suppliers now rated non-investment grade**

vs. <5% in MedTech & industrials — refinancing risk is structural.

✓ VERIFIED

86–94%**Of operational spreadsheets carry at least one error**

~1% of formula cells wrong in field audits; humans miss most.

✓ VERIFIED

Feb 2027**EU digital battery passport becomes mandatory**

18 Feb 2027 — a hard deadline the whole industry is racing.

✓ VERIFIED

Cash-starved and mandate-loaded at once — the canonical condition where a low-cost, high-ROI compliance tool finds demand. No compliant PPAP package → no purchase order.

02 The gap — measured, not asserted

Scored against a committed 37-case benchmark

Spreadsheets

BROAD BUT UNGUARDED

Touch every document, but passively accept whatever is typed. No engine to catch what's wrong — structurally incapable of every coherence, genealogy and provenance case, regardless of operator skill.

Point solutions

DEEP BUT NARROW

Nail one task — SPC statistics, say — and leave everything else back in the spreadsheet. Even granting perfect in-scope accuracy, they still can't cover most of the job.

The union no one covers

THE ARCHITECTURE-LEVEL OPENING

- + Cross-document integrity
- + Quality-plan (PPAP/FMEA/control-plan) coherence
- + Recall genealogy
- + Tamper-evident records
- + Active guardrails on bad data

Covering that union is an **architecture-level job** — not a feature either incumbent can bolt on. A passive spreadsheet has no engine for coherence, no graph for genealogy, no signing for provenance. That is the opening SQ-OS is built for.

03 The offering — one system of record for the whole job

Compliance core

PPAP, APQP, FMEA, control plans and SPC — assembled, cross-checked and kept consistent, priced for single-digit-margin shops.

Active guardrails

Catches the errors a spreadsheet waves through — stale revisions, mis-used risk scores, out-of-spec capability claims.

Traceability spine

Part-level serialization and genealogy: turn a total recall into a targeted one, from raw lot to finished serial.

Battery-passport ready

The same data spine produces the EU digital battery passport every EV supplier needs by February 2027.

04 Why believe it — define the bar, then measure the gap

Most compliance software asks you to trust a claim. We built a **testable benchmark of what "correct" means** for automotive compliance — grounded in the AIAG / IATF standards and the EU battery regulation — and committed it cryptographically **before** building the product, a direct guard against optimizing to a moving target. Every check traces to a published standard; where a number can't be verified, we don't publish one. The moat compounds: switching cost once we hold the quality record of record, plus a **mandate stack that only grows**.

PPAP

AIAG-VDA FMEA

IATF 16949

EU 2023/1542

Crypto provenance

Genealogy graph

MEASURED, NOT ASSERTED**19/37**cases a spreadsheet binder is **structurally incapable** of**29/37**

cases an SPC point solution can't cover — even granted perfect accuracy

Coverage over the committed corpus of 37 scored cases. A companion IEEE whitepaper documents the benchmark and the measured incumbent gap.

The bar is defined. The gap is measured. The clock is running.

We're selecting design partners — mid-tier suppliers with acute, mandated compliance pain — to prove the wedge together, then attach the traceability spine and battery passport ahead of the 2027 deadline.

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BUILD WHAT DOESN'T EXIST YET

Sources: Roland Berger / Lazard, *Global Automotive Supplier Study 2025* (4.7% EBIT, 2024, down from 5.3%; >40% of top-25 suppliers non-investment grade). · Regulation (EU) 2023/1542, Arts. 77–78 & Annex XIII (battery passport mandatory 18 Feb 2027). · R. Panko, field audits of operational spreadsheets (~1% cell-error rate; 86–94% of spreadsheets contain ≥1 error). · Benchmark coverage figures are from SQ-OS's own committed 37-case corpus. Thesis-stage: benchmark and IEEE whitepaper complete; product and pricing are the thesis under test.